GUITAR EFFECTS PEDAL — STM32F411 + PCM1808 ADC + PCM5102A DAC

Signal path: guitar -> INPUT buffer -> ADC (PCM1808) -> I2S -> MCU (overdrive, tremolo, delay) -> I2S -> DAC (PCM5102A) -> OUTPUT buffer -> amp
Every wire is drawn as a label: two labels with the same name are the same wire.
Power: 9 V centre-negative adapter. Net names: +9V (after the protection diode), +5V, +5VA (ADC analog), +3V3A (converters), +3V3_MCU (from the Black Pill), VBIAS (half of 9 V, the op-amps' 'zero').

POWER

9 V in -> protection -> 5 V, 3.3 V and VBIAS (4.3 V)

File: power.kicad_sch

INPUT

guitar -> 1 MOhm buffer (x1.5) -> anti-alias filter -> ADC

File: input.kicad_sch

ADC

PCM1808 turns the analog signal into numbers (I2S)

File: adc.kicad_sch

DAC

PCM5102A turns numbers back into an analog signal

File: dac.kicad_sch

OUTPUT

DAC -> filter -> attenuator -> buffer -> output jack

File: output.kicad_sch

MCU

STM32F411 Black Pill runs the effects

File: mcu.kicad_sch

CONTROLS

6 knobs, 3 footswitches, 3 LEDs

File: controls.kicad_sch

TEST

TEST POINTS and mounting holes

File: test.kicad_sch

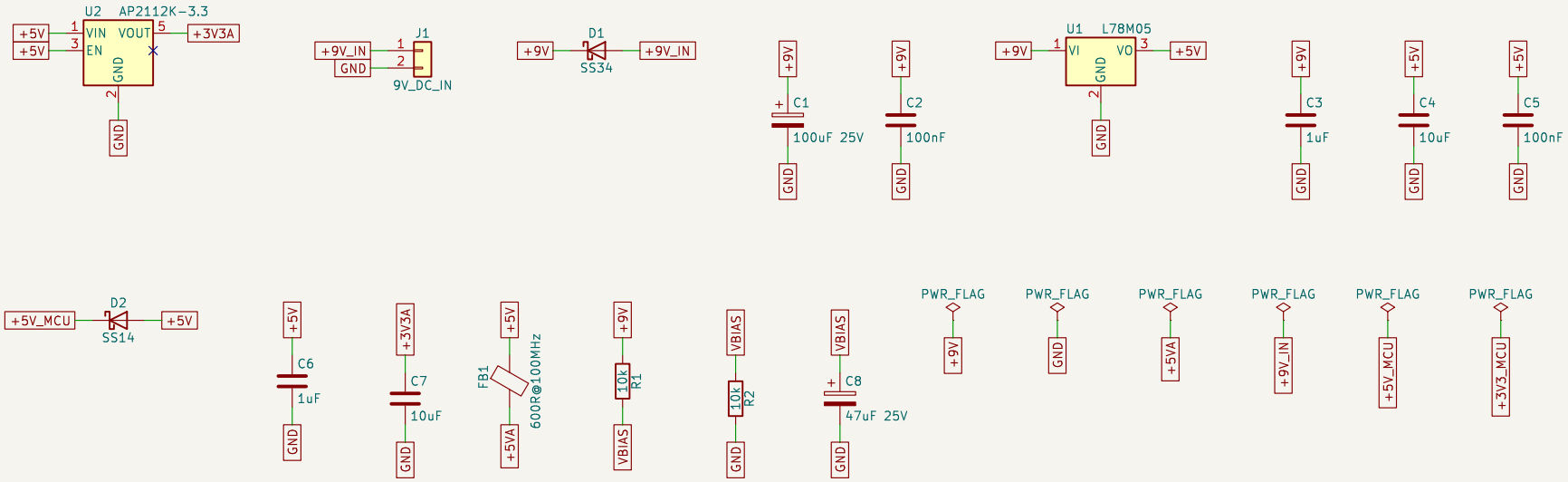
Embedded audio learning project

Sheet: /
File: pedal.kicad_sch

Title: Guitar effects pedal

Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/9

POWER: 9 V in -> protection -> 5 V, 3.3 V and VBIAS (4.3 V)



J1 (9V_DC_IN): Wire to a 2.1 mm DC jack: sleeve -> pin 1 (+9V), centre pin -> pin 2 (GND). Pedal standard = centre negative.
D1 (SS34): Reverse polarity protection: blocks current if the adapter is wrong
C1 (100uF 25V): Bulk supply reservoir
C2, C5 (100nF): Ceramic capacitor
U1 (L78M05): 9 V -> 5 V regulator, 500 mA
C3 (1uF): Regulator input cap
C4 (10uF): Regulator output cap
D2 (SS14): Feeds the Black Pill 5V pin. Stops USB power flowing back into our 5 V rail
U2 (AP2112K-3.3): 5 V -> 3.3 V clean supply for the audio converters
C6 (1uF): Ceramic capacitor
C7 (10uF): Ceramic capacitor
FB1 (600R@100MHz): Ferrite bead: keeps digital noise off the ADC's analog 5 V
R1 (10k): Bias divider top
R2 (10k): Bias divider bottom
C8 (47uF 25V): Makes VBIAS a quiet 'AC ground' at half supply

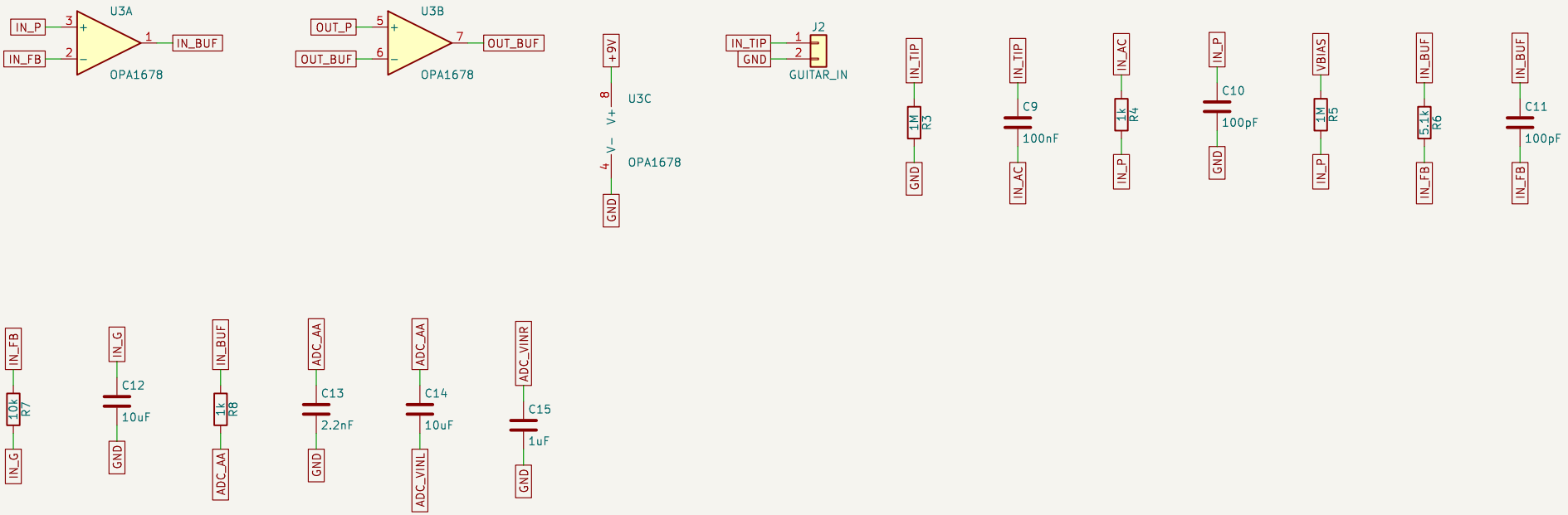
Embedded audio learning project

Sheet: /POWER/
File: power.kicad_sch

Title: Guitar pedal — power

Size: A4	Date:	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/9

INPUT: guitar -> 1 MOhm buffer (x1.5) -> anti-alias filter -> ADC



J2 (GUITAR_IN): Wire to the input jack: tip -> 1, sleeve -> 2
R3 (1M): Pull-down: stops a 'pop' when you plug in
C9 (100nF): Blocks DC from the guitar side
R4 (1k): With C10: filters radio interference
C10 (100pF): Ceramic capacitor
R5 (1M): Biases the op-amp input to VBIAS; 1 MOhm = what a guitar wants to see
U3 (OPA1678): Dual audio op-amp, rail-to-rail output. NE5532 fits too (same pins, works on 9 V)
R6 (5.1k): Gain = 1 + R6/R7 = 1.51 (+3.6 dB)
C11 (100pF): Rolls off gain above 300 kHz (stability)
R7 (10k): Resistor 0805 1%
C12 (10uF): Gain only for AC; at DC the stage is a follower and stays at VBIAS
R8 (1k): Anti-alias filter R (72 kHz with C13)
C13 (2.2nF): Ceramic capacitor
C14 (10uF): Blocks VBIAS (4.3 V) from the ADC, which centres itself at 2.5 V
C15 (1uF): Unused right ADC input: AC-grounded so it stays quiet

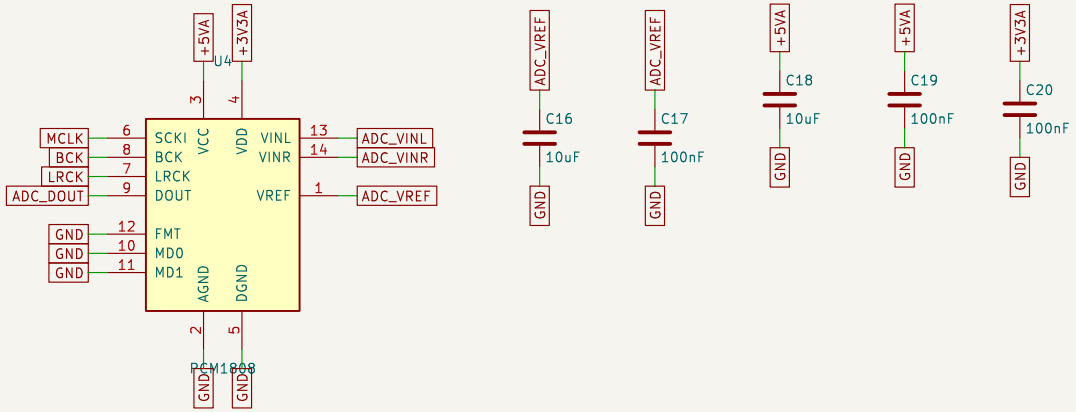
Embedded audio learning project

Sheet: /INPUT/
File: input.kicad_sch

Title: Guitar pedal — input

Size: A4	Date:	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/9

ADC: PCM1808 turns the analog signal into numbers (I2S)



U4 (PCM1808): 24-bit stereo ADC. MD0=MD1=0: slave mode. FMT=0: I2S
C16 (10uF): ADC reference decoupling (datasheet: 10uF + 0.1uF)
C17, C19, C20 (100nF): Ceramic capacitor
C18 (10uF): Ceramic capacitor

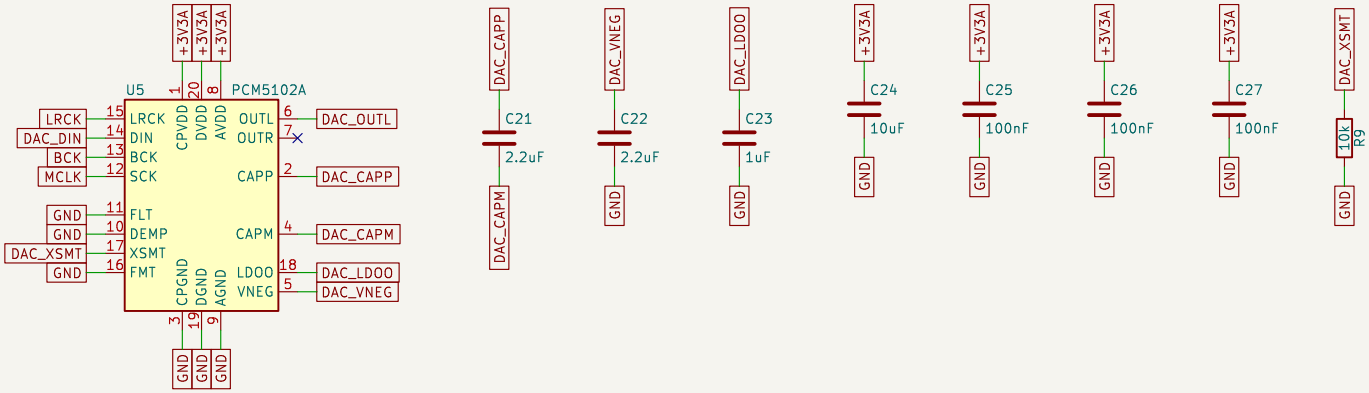
Embedded audio learning project

Sheet: /ADC/
File: adc.kicad_sch

Title: Guitar pedal – adc

Size: A4	Date:	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/9

DAC: PCM5102A turns numbers back into an analog signal



U5 (PCM5102A): 32-bit stereo DAC with built-in negative charge pump: output centred on 0 V
C21 (2.2uF): Charge-pump flying cap
C22 (2.2uF): Charge-pump output (-3.3 V)
C23 (1uF): Internal 1.8 V regulator decoupling
C24 (10uF): Ceramic capacitor
C25 (100nF): AVDD decoupling
C26 (100nF): CPVDD decoupling
C27 (100nF): DVDD decoupling
R9 (10k): Keeps the DAC muted until the MCU says go (no power-on pop)

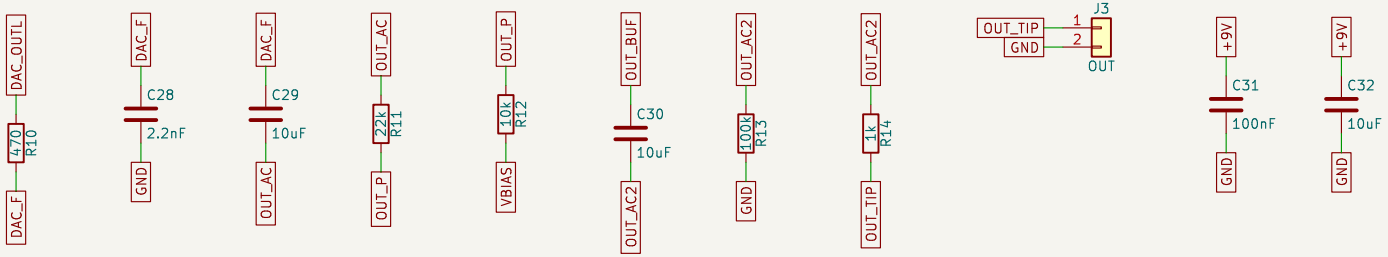
Embedded audio learning project

Sheet: /DAC/
File: dac.kicad_sch

Title: Guitar pedal – dac

Size: A4	Date:	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/9

OUTPUT: DAC -> filter -> attenuator -> buffer -> output jack



R10 (470): DAC output filter (datasheet: 470 Ohm + 2.2 nF)
C28 (2.2nF): Ceramic capacitor
C29 (10uF): Blocks DC between the 0 V DAC and the 4.3 V op-amp
R11 (22k): Attenuator: 10k/(22k+10k) = 0.31, full scale 3 V peak -> 0.93 V peak
R12 (10k): Bottom of attenuator, also biases the op-amp to VBIAS
C30 (10uF): Blocks VBIAS from the output jack
R13 (100k): Pull-down: keeps the jack at 0 V DC (no pop)
R14 (1k): Protects the op-amp from shorts and long cables
J3 (OUT): Wire to the output jack: tip -> 1, sleeve -> 2
C31 (100nF): Op-amp supply decoupling
C32 (10uF): Ceramic capacitor

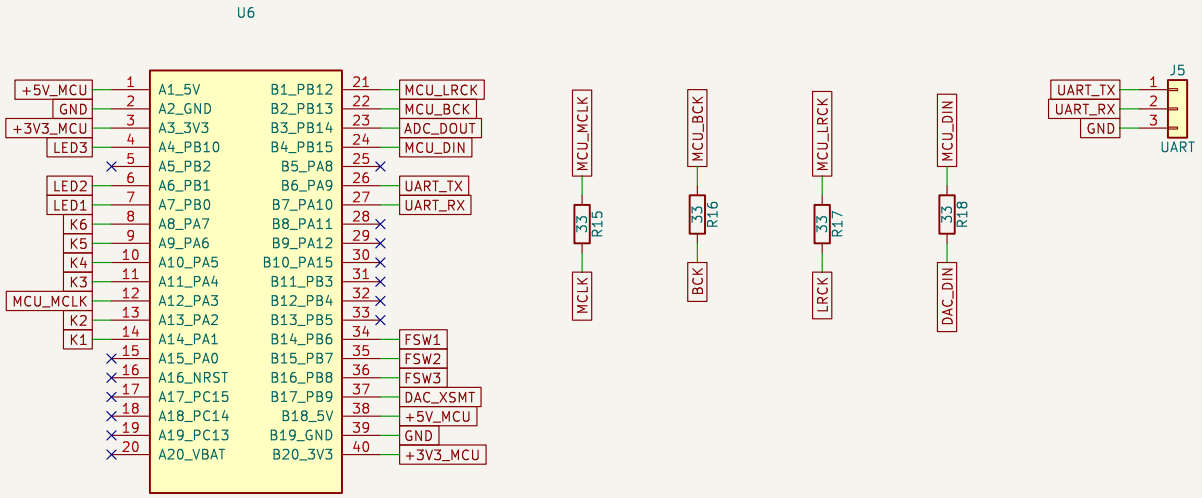
Embedded audio learning project

Sheet: /OUTPUT/
File: output.kicad_sch

Title: Guitar pedal – output

Size: A4	Date:	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/9

MCU: STM32F411 Black Pill runs the effects



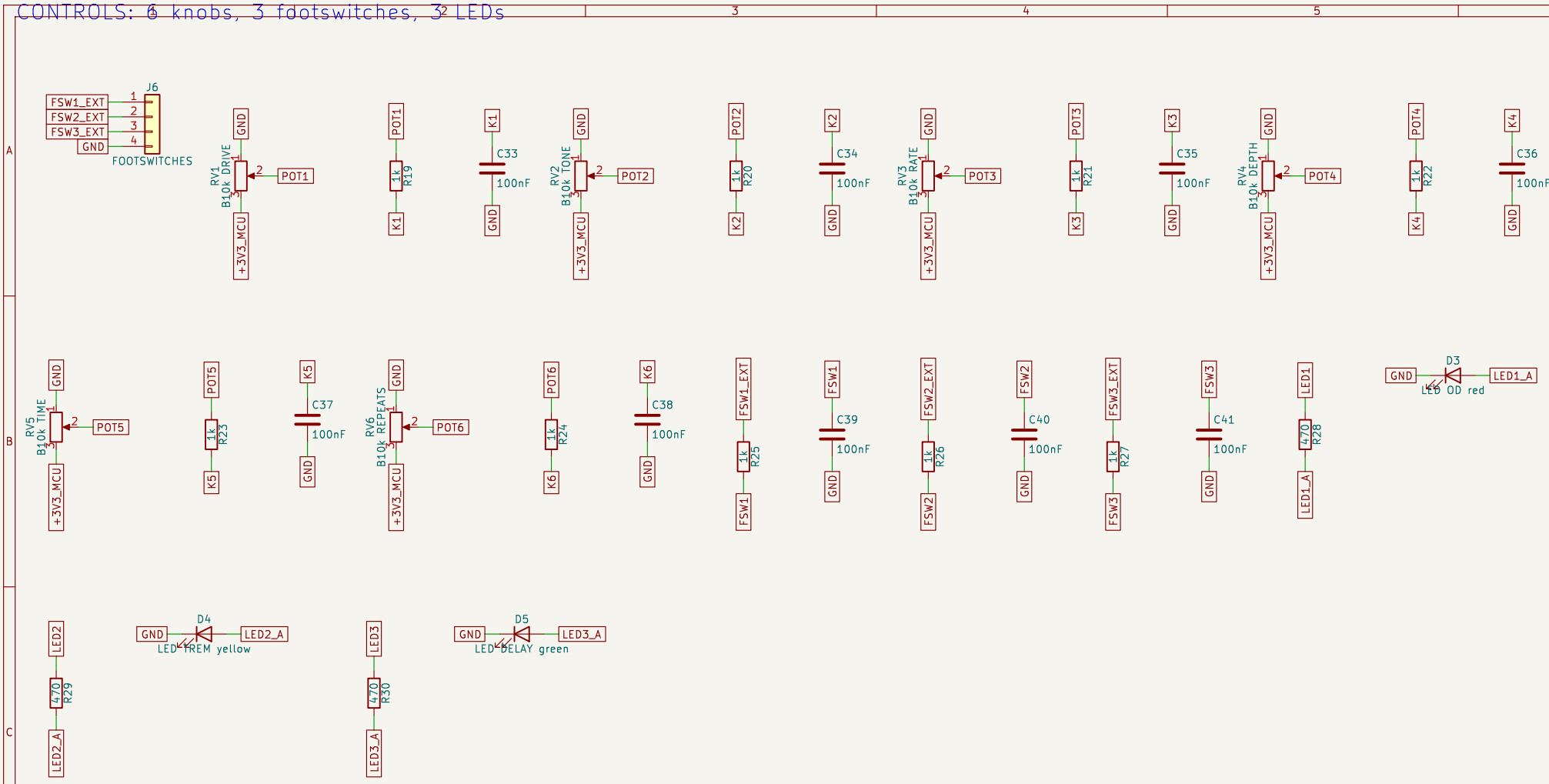
U6 (WeAct BlackPill STM32F411CEU6): Plugs into two 1x20 female headers. Not assembled by the factory — you buy and plug it
R15 (33): Series resistors on fast clock lines: soften edges, less ringing/noise
R16, R17, R18 (33): Resistor 0805 1%
J5 (UART): Debug serial port: connect a USB-serial adapter (its RX to pin 1)

Embedded audio learning project

Sheet: /MCU/
File: mcu.kicad_sch

Title: Guitar pedal — mcu

Size: A4	Date:	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/9



RV1 (B10k DRIVE): Knob DRIVE. CCW end to GND, CW end to 3.3 V
R19, R20, R21, R22, R23, R24 (1k): Knob filter R
C33, C34, C35, C36, C37, C38 (100nF): Knob filter C: steadies the ADC reading
RV2 (B10k TONE): Knob TONE. CCW end to GND, CW end to 3.3 V
RV3 (B10k RATE): Knob RATE. CCW end to GND, CW end to 3.3 V
RV4 (B10k DEPTH): Knob DEPTH. CCW end to GND, CW end to 3.3 V
RV5 (B10k TIME): Knob TIME. CCW end to GND, CW end to 3.3 V
RV6 (B10k REPEATS): Knob REPEATS. CCW end to GND, CW end to 3.3 V
J6 (FOOTSWITCHES): Three momentary (soft-touch) SPST footswitches: each one between pin 1/2/3 and pin 4 (GND)
R25, R26, R27 (1k): Protects the MCU pin from static on the switch wire
C39, C40, C41 (100nF): Hardware debounce + noise filter
R28, R29, R30 (470): LED current limit: (3.3-2.0)/470 = 2.8 mA
D3 (LED OD red): Status LED for OD
D4 (LED TREM yellow): Status LED for TREM
D5 (LED DELAY green): Status LED for DELAY

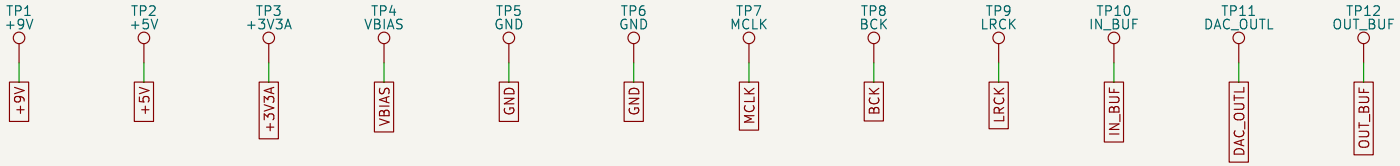
Embedded audio learning project

Sheet: /CONTROLS/
File: controls.kicad_sch

Title: Guitar pedal – controls

Size: A4	Date:	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/9

TEST POINTS and mounting holes



TP1 (+9V): Measure here with the multimeter or scope
TP2 (+5V): Measure here with the multimeter or scope
TP3 (+3V3A): Measure here with the multimeter or scope
TP4 (VBIAS): Measure here with the multimeter or scope
TP5, TP6 (GND): Measure here with the multimeter or scope
TP7 (MCLK): Measure here with the multimeter or scope
TP8 (BCK): Measure here with the multimeter or scope
TP9 (LRCK): Measure here with the multimeter or scope
TP10 (IN_BUF): Measure here with the multimeter or scope
TP11 (DAC_OUTL): Measure here with the multimeter or scope
TP12 (OUT_BUF): Measure here with the multimeter or scope

Embedded audio learning project

Sheet: /TEST/
File: test.kicad_sch

Title: Guitar pedal – test

Size: A4	Date:	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 9/9